

Chapter 9

Systematic Uncertainties

Systematic uncertainties quantify the limited knowledge of the detector response and of the physics processes used to predict the signal and background yields. They are evaluated by varying calibration corrections, efficiency scale factors and simulation parameters, or by comparing the nominal prediction with an alternative calculation. These variations can change both the total event rate and the kinematic distributions, and therefore the fraction of events passing each region selection. For the single-bin regions used in this analysis, their effects enter the fit through the resulting changes in the selected yields.

Experimental uncertainties are discussed first, followed by uncertainties in the theoretical modelling of the backgrounds and signal. The nominal MC configurations and object calibrations are given in Chapters 5 and 6; the discussion here concentrates on their uncertainties and propagation. In particular, the normalisations of $W + \text{jets}$ and top-quark backgrounds are constrained by the CRs introduced in Chapter 8, so their modelling uncertainties must describe the extrapolation from those CRs to the SRs and VRs. Each independent source is represented in the fit by a nuisance parameter, which allows the prediction to vary within the uncertainty assigned to that source. The likelihood and the constraints on these parameters are described in Chapter 10.

9.1 Experimental uncertainties

Experimental uncertainties include the luminosity measurement, the pile-up distribution and the calibration and selection efficiencies of reconstructed objects. An efficiency uncertainty is propagated by varying the corresponding scale factor in the simulated event weight. An energy or momentum uncertainty instead changes the object kinematics, after which the event variables and selections are reevaluated. The latter can move events across a selection threshold or between regions. For example, a jet-energy variation can change both the leading-jet p_T used to

separate the resonant and non-resonant selections and the reconstructed E_T^{miss} . Variations from the same calibration source are propagated coherently wherever that source applies, retaining its prescribed dependence on object kinematics and type.

9.1.1 Luminosity and pile-up reweighting

The 0.83% luminosity uncertainty quoted in Section 5.1 is propagated as a common normalisation uncertainty for predictions normalised using the integrated luminosity [1]. For $W + \text{jets}$ and top-quark backgrounds, a common change in their simulated normalisation can be absorbed by the freely fitted normalisation factors. The luminosity uncertainty therefore acts directly on the signal and on background contributions whose normalisations are taken from simulation.

Pile-up reweighting corrects the simulated distribution of the mean number of interactions per bunch crossing, μ , to reproduce that in data. The data-derived μ profile used to calculate these weights is rescaled to improve the agreement in the reconstructed vertex multiplicity. The nominal rescaling factor is $1/1.03$; alternative factors of $1/1.07$ and $1/0.99$ are used to derive the downward and upward variations. The resulting event weights are propagated through the analysis selections. This uncertainty concerns the frequency of pile-up interactions, whereas the residual effect of pile-up on calibrated jets is included separately in the jet uncertainties below.

9.1.2 Tau energy scale and efficiency calibration

The TES calibration in Section 6.7 sets the nominal visible tau momentum. Its uncertainty is constrained by in situ measurements with $Z \rightarrow \tau\tau$ events, supplemented by detector-response studies using simulation and test-beam information at high p_T [2–4]. The in situ correction to the simulated tau energy is typically below 1%, with an uncertainty in the energy scale below 1.5%. These percentages refer to the energy calibration; the change in a selected event yield depends on the spectrum near the analysis thresholds.

Five TES components are propagated separately. Two describe the experimental uncertainties and fit uncertainty of the in situ determination. The remaining components describe detector modelling, the choice of hadronic-interaction model in the detector simulation, and calibration non-closure. Here, non-closure is the residual difference between the calibrated response and its reference value when the calibration is tested on simulation. The detector and hadronic-interaction components retain sensitivity to the response at tau momenta beyond the main in situ calibration range. Each TES variation changes the tau four-momentum and is propagated to $\vec{p}_T^{\text{miss}}$, m_T and the jet–tau invariant mass before the region requirements are applied.

Separate uncertainties are included for tau reconstruction, RNN identification and electron rejection. The reconstruction and identification efficiencies are checked in $t\bar{t}$ data, and the corresponding scale factors depend on tau p_T and prong multiplicity. For the Tight identification working point, the scale factors are close to unity, with uncertainties of 5–8% for one-prong candidates and 6–11% for three-prong candidates. The identification uncertainties are represented by independent components for the two prong multiplicities and for the calibration p_T bins, rather than by one common uncertainty for all tau candidates. The bins have boundaries at 20, 25, 30, 40, 60, 80 and 120 GeV, with the last bin extending above 120 GeV.

Electron-rejection uncertainties distinguish the efficiency for a hadronically decaying tau from the probability for an electron to pass the tau selection. The former has its own efficiency uncertainty, while statistical and systematic components are provided for the latter. The scale-factor convention used for the electron veto is stated in Section 6.7. Uncertainties in jets misidentified as tau candidates are treated separately in Section 9.2.4.

9.1.3 Jets and missing transverse momentum

The JES and JER uncertainties are derived from the calibration and balance measurements outlined in Section 6.5 [5]. They depend on jet p_T and η , and include both the precision of the measurements and differences between the measured and simulated responses. The JES uncertainty is organised into components with different physical origins:

- **In situ calibration uncertainties** include the statistical precision of the balance measurements, the calibration of the reference objects and the modelling of additional radiation and event selections. The η intercalibration also includes the uncertainty in transferring the central-jet calibration to other detector regions.
- **Pile-up uncertainties** describe residual response changes with μ , the number of reconstructed vertices, jet p_T and the event topology used to estimate the pile-up energy density.
- **Flavour uncertainties** include the different responses to quark- and gluon-initiated jets and the uncertainty in their fractions in a sample. A dedicated component describes the response to b -jets.
- **High-energy response uncertainties** include the single-particle response used beyond the reach of the balance measurements and the modelling of jets whose showers extend into the muon system.

The CategoryReduction scheme combines calibration components while approximately preserving their correlations across jet kinematics. The conservative default flavour-response and flavour-composition uncertainties are used, without a sample-specific reduction.

The Run 2 samples are reconstructed with Release 22, while parts of the in situ jet calibration originate from Release 21 measurements. Additional uncertainties cover changes in calorimeter noise thresholds and vertex reconstruction, together with residual non-closure when the earlier in situ calibration is applied to the reprocessed samples. For the ATLFast3 signal samples, further JES and JER components cover response and resolution differences from the full detector simulation used for the backgrounds. Their effect on the selected signal yield is approximately 1%.

The JER uncertainty changes the width of the jet-response distribution and is propagated through momentum smearing. The FullJER prescription is used to retain the correlations of the resolution uncertainties. It combines variations obtained by smearing simulation and pseudo-data, so that a component that changes sign across jet kinematics retains the corresponding anticorrelation. The two sets of smearing variations are ingredients of a common JER variation, rather than independent uncertainty sources. The efficiency scale factor for NNJVT has a separate uncertainty, which changes the probability for a hard-scattering jet to pass the pile-up rejection requirement.

All hard-object momentum variations are propagated to $\vec{p}_T^{\text{miss}}$ using its definition in Eq. (6.3). The track soft term has three additional components: a scale uncertainty along the hard-term momentum and resolution uncertainties parallel and perpendicular to it [6]. They are obtained from the balance between the hard and soft terms in $Z \rightarrow ee$ events, where little missing momentum is expected from the hard process. Differences between data and simulation in the mean parallel projection determine the scale uncertainty, while differences in the widths of the two projections determine the resolution uncertainties. The soft-term variations are evaluated independently of the hard-object variations, and both enter the uncertainty model for E_T^{miss} and the variables constructed from it.

9.1.4 Jet flavour-tagging efficiency

The flavour-dependent scale factors in Section 6.6 correct the probabilities for jets to pass or fail the tagging requirement. Their uncertainties include the statistical precision of the calibration samples and systematic effects from background subtraction, object calibration and process modelling [7–9]. Uncertainties are evaluated separately for the b -jet tagging efficiency and for the c -jet and light-jet mistagging efficiencies. They affect both SR1b and SR0b, since uncertainties in the probabilities to tag or veto a jet change the predicted yields in both regions.

Correlations between the calibration bins are represented using an eigenvector decomposition of their covariance matrix. Each retained eigenvector gives a coherent variation of the scale factors across the bins, while different eigenvectors are treated as independent nuisance parameters. The Medium reduction scheme is used for the GN2v01 calibration. This calibration uncertainty describes the detector's ability to identify a given jet flavour. The uncertainty in the amount of heavy-flavour production in $W + \text{jets}$ events is evaluated separately in Section 9.2.2.

9.1.5 Electrons and muons

Electron energy-scale and resolution uncertainties are propagated by varying the calibration described in Section 6.3 [10]. The efficiency corrections have separate reconstruction, identification, isolation and trigger uncertainties [11, 12]. They are evaluated for the working points used in the analysis and propagated through the simulated event weights.

Muon momentum uncertainties include the overall scale and resolution and residual charge-dependent curvature biases from detector alignment [13]. The latter are commonly expressed as sagitta uncertainties, where the sagitta measures the bending of a track in the magnetic field. Their components include the statistical precision of the calibration, residual calibration bias and extrapolation to high p_T . Muon efficiency uncertainties cover reconstruction, isolation, the track-to-vertex requirements, trigger selection and the rejection of poorly measured muons [14, 15]. Statistical and systematic components are retained separately where supplied by the calibration.

These uncertainties act mainly in CRs and VRs containing a selected electron or muon and are smaller than the tau uncertainties in this analysis. They can nevertheless change the background normalisations inferred from the CRs and hence the predictions in the SRs. The trigger choices and their efficiency corrections are described in Section 5.4.

9.2 Theoretical uncertainties

Theoretical modelling uncertainties arise from the finite perturbative accuracy of the hard-scattering calculation, the proton PDFs and the treatment of radiation and hadronisation. The nominal generators and PDF sets are given in Sections 5.2 and 5.3. Here, each variation is used to determine how the selected prediction changes when one part of that modelling is altered.

9.2.1 Uncertainties in the transfer from control regions

For a background b constrained in a CR, the simulated transfer factor to a target region R is defined as

$$T_b^R = \frac{N_b^{R,\text{nom}}}{N_b^{\text{CR},\text{nom}}}, \quad (9.1)$$

where R denotes an SR or VR, and $N_b^{R,\text{nom}}$ is the nominal simulated yield after the corresponding selection. The appropriate CR is CRW for $W + \text{jets}$ and CRTop for the combined $t\bar{t}$ and single-top background, within the same resonant or non-resonant selection. As described in Section 8.1, the fitted factor λ_b scales the nominal prediction in both regions. The predicted target yield can therefore be written as either $\lambda_b N_b^{R,\text{nom}}$ or $T_b^R \lambda_b N_b^{\text{CR},\text{nom}}$. The fitted CR contribution includes the treatment of other processes in that region; it is not identified with the raw observed CR count.

A modelling variation must be evaluated in both the target region and its CR. Its fractional effect on the transfer factor is

$$\begin{aligned} \delta_b^R &\equiv \frac{T_b^{R,\text{var}}}{T_b^{R,\text{nom}}} - 1 \\ &= \frac{N_b^{R,\text{var}}/N_b^{R,\text{nom}}}{N_b^{\text{CR},\text{var}}/N_b^{\text{CR},\text{nom}}} - 1. \end{aligned} \quad (9.2)$$

The double ratio removes changes common to the CR and target region. For example, a variation that multiplies both yields by the same factor changes the inclusive normalisation but leaves the transfer factor unchanged. A residual uncertainty remains when the variation changes the acceptance differently in the two regions. This is why the theoretical uncertainty in the inclusive cross-section is not applied as an additional independent normalisation constraint on these freely normalised backgrounds.

For one uncertainty source, its effect can be represented by a dimensionless modifier $\kappa_b^R(\theta)$, with $\kappa_b^R(0) = 1$, such that

$$N_b^{R,\text{pred}}(\theta) = \lambda_b N_b^{R,\text{nom}} \kappa_b^R(\theta), \quad N_b^{\text{CR},\text{pred}} = \lambda_b N_b^{\text{CR},\text{nom}}. \quad (9.3)$$

At the evaluated variation, $\kappa_b^R = 1 + \delta_b^R$. Equation (9.3) isolates the theoretical transfer uncertainty; the experimental and MC statistical uncertainties are included separately. The theoretical transfer variations for the dominant backgrounds are applied to their SR and VR predictions, while their overall rates remain constrained by the CR data. The small cross-contamination of

top events in CRW and $W + \text{jets}$ events in CRTop is not assigned a separate transfer uncertainty of this type.

9.2.2 $W + \text{jets}$ modelling

QCD scales The renormalisation scale μ_R specifies the scale at which the strong coupling is evaluated, while the factorisation scale μ_F separates the long-distance proton structure included in the PDFs from the hard-scattering calculation. An all-orders prediction would not depend on these arbitrary choices, but a calculation truncated at fixed order retains a residual dependence. Scale variations are therefore used to estimate the effect of missing higher-order QCD contributions.

The internal event weights in the SHERPA samples provide seven combinations of scale multipliers,

$$(\mu_R/\mu_R^0, \mu_F/\mu_F^0) \in \{(1, 1), (1/2, 1/2), (1/2, 1), (1, 1/2), (2, 2), (2, 1), (1, 2)\}, \quad (9.4)$$

where the superscript 0 denotes the nominal scale choice. The two combinations in which one scale is doubled and the other halved are excluded. Coherent variations of the matrix-element and parton-shower scales are used [16]. For each combination, the target-to-CR double ratio is calculated before taking the envelope around the nominal transfer factor. This preserves the cancellation between regions for the same physical variation.

PDFs and the strong coupling PDF uncertainties describe the limited knowledge of the parton momentum distributions inferred from experimental data [17, 18]. They are propagated using the NNPDF3.0nnlo Hessian variations provided with the samples. In this representation, independent variations along the fitted PDF error directions are combined using the corresponding Hessian prescription. The central predictions of the MMHT2014nnlo and CT14nnlo sets are also compared with the nominal prediction, and dedicated NNPDF variations probe the uncertainty in α_s . All comparisons use the same variation in the CR and target region when constructing Eq. (9.2).

Electroweak corrections The SHERPA samples provide event weights for approximate NLO electroweak corrections [16]. These corrections become more significant at high momentum and can change the extrapolation into the selected tails. Their effect on the transfer factor is included as an additional uncertainty by comparing the electroweak-reweighted and nominal predictions. The resulting variations are approximately 4% and 1% for the resonant and non-resonant SR1b selections, respectively.

Heavy-flavour fraction The b -vetoed CRW constrains a different flavour mixture from the tagged SRs and VRs. Measurements and modelling studies of vector-boson production with heavy-flavour jets show that the simulated flavour fractions and normalisations can differ from data by tens of percent, with a dependence on the selected phase space [19, 16, 20, 21]. Following the validation discussed in Section 8.5.1, an additional 30% uncertainty is assigned to the transfer from W production with light-flavour hadrons to production with heavy-flavour hadrons. It is implemented as an overall variation of the $W + \text{jets}$ yield in regions with one or two tagged jets. Independent nuisance parameters are used for the resonant and non-resonant selections, while the tagged regions within each selection share the corresponding parameter. This additional uncertainty is not applied directly to SR0b or the b -vetoed CRW. It expresses uncertainty in the flavour composition used for the extrapolation; it does not vary the tagging efficiency of individual jets.

9.2.3 Top-quark modelling

The top-quark prediction combines $t\bar{t}$ and single-top production with the common normalisation described in Section 8.3. The following variations probe the modelling of this combined prediction [22]:

- **NLO matching** is tested by comparing the nominal POWHEG+PYTHIA prediction with an alternative sample using `pthard=1` in PYTHIA. This setting changes the definition of the shower-emission veto used to match the parton shower to the radiation generated by POWHEG. The comparison therefore probes the matching prescription while retaining the same hard-process generator.
- **Parton shower and hadronisation** are varied by replacing PYTHIA with HERWIG in samples generated with POWHEG. Changes in the additional radiation and jet structure can alter both the jet-multiplicity acceptance and the high-momentum tails.
- **Renormalisation and factorisation scales** are varied using the internal event weights. Their effects are evaluated on the transfer factors, with separate variations retained for μ_R and μ_F .
- **Initial-state radiation** is varied using the A14 Var3c weights, which change the strength of the initial-state shower radiation. **Final-state radiation** is varied through the shower renormalisation scale, using factors of two and one half. These variations probe the amount and kinematics of additional radiation and therefore the efficiency of the jet selections.

- **PDF and α_s uncertainties** are propagated through the available generator weights using the same target-to-CR comparison as for $W + \text{jets}$.
- **The h_{damp} parameter** controls the damping of hard real-emission contributions that are resummed by POWHEG. An alternative sample with $h_{\text{damp}} = 3m_t$ is compared with the nominal $h_{\text{damp}} = 1.5m_t$ setting, where m_t is the top-quark mass.

Three further variations address specific features of the $t\bar{t}$ and Wt components. First, the nominal NLO $t\bar{t}$ sample is reweighted to the NNLO prediction for the top-quark p_T distribution [23]. The difference is used as a modelling uncertainty in the shape of that distribution and its effect on the selected yields. It is distinct from the inclusive NNLO+NNLL normalisation already specified in Section 5.3. Second, an alternative shower recoil prescription is used for $t\bar{t}$, in which the recoil from radiation in the top decay is assigned to the top quark rather than according to the nominal colour-based prescription. This tests how momentum conservation in the shower changes the reconstructed top-decay kinematics.

Third, Wt production overlaps with $t\bar{t}$ production beyond LO because some real-emission diagrams contain a second resonant top quark [24]. The nominal diagram-removal prescription omits these doubly resonant diagrams from the Wt calculation. The alternative diagram-subtraction prescription subtracts a local approximation to the resonant $t\bar{t}$ contribution at the cross-section level, retaining interference terms. Their difference estimates the uncertainty in separating the two contributions. This variation is evaluated using the Wt samples and then propagated to the combined top-quark transfer factor. Likewise, the NNLO reweighting and recoil variations act on the $t\bar{t}$ component before the change in the total top prediction is evaluated.

9.2.4 $Z \rightarrow \nu\nu$ with a jet misidentified as a tau

The $Z \rightarrow \nu\nu + \text{jets}$ background with a jet passing the tau identification has a dedicated 100% normalisation uncertainty. Its origin is the $Z \rightarrow \ell\ell$ validation study in Section 8.4.1, where differences of up to 60% are observed in the total event yield while the kinematic shapes are generally described by simulation. These differences include both the modelling of $Z + \text{jets}$ production and the probability for a jet to pass the tau selection. The conservative uncertainty is propagated to the simulated fake-tau contribution, whose small yield limits its effect on the total SR background.

9.2.5 Other background components

Diboson production and the remaining $Z + \text{jets}$ backgrounds retain their nominal simulated yields. Conservative 30% normalisation uncertainties are assigned to these contributions in the SRs and VRs. These are analysis uncertainties for the selected event yields, rather than measurements of the precision of the inclusive production cross-sections. The dedicated fake-tau uncertainty in Section 9.2.4 is retained for the corresponding misidentified-tau contribution.

9.2.6 Signal modelling

The signal prediction is varied through the renormalisation and factorisation scales and the NNPDF3.0nlo PDF uncertainties. The variations are evaluated for the generated mass and coupling configurations described in Section 5.2. Unlike the dominant backgrounds, the signal is not normalised using a CR measurement, so changes in both its production rate and selection acceptance contribute to its uncertainty. The resulting theory uncertainty is typically of order 10–20%, with a dependence on the signal parameters and the selected region. The signal weights used for this evaluation do not provide a separate pair of α_s variations for the baseline PDF set, so no independent signal α_s uncertainty is derived from them. The interference contribution defined in Eq. (5.4) remains a separate signed prediction; it is not part of the background transfer-factor construction.

9.2.7 Evaluation in regions with limited MC statistics

The high- $m_{b\tau}$ and high- E_T^{miss} requirements leave few top-quark events in some of the alternative samples. A direct comparison after the complete selection can then be dominated by fluctuations in the simulated event weights, rather than by the modelling change being tested. For these sources, the uncertainty is evaluated using distributions before the final threshold and, where necessary, a relaxed selection.

The alternative sample is first normalised to the nominal yield in its dedicated CR. An $N - 1$ distribution is then formed by retaining the region selection except for the threshold on the variable being studied: $m_{b\tau}$ for the resonant SR1b or E_T^{miss} for the non-resonant SR1b. For bin i of that distribution, the ratio after CR normalisation is

$$r_i = \frac{N_i^{\text{var}}}{N_i^{\text{nom}}} \frac{N^{\text{CR,nom}}}{N^{\text{CR,var}}}. \quad (9.5)$$

The ratio is smoothed with the 353QH twice algorithm implemented in ROOT [25, 26]. The smoothed ratio $\tilde{r}_i$ is applied to the nominal distribution, and the bins passing the final selection

are summed:

$$1 + \delta_b^R = \frac{\sum_{i \in R} \tilde{r}_i N_i^{\text{nom}}}{\sum_{i \in R} N_i^{\text{nom}}}. \quad (9.6)$$

This gives a change in the selected transfer factor while reducing the influence of isolated fluctuations in the alternative sample.

When the available statistics are insufficient even before smoothing, the jet-multiplicity requirement is removed and $N_b = 1$ is replaced by $N_b \geq 1$ for the SR uncertainty evaluation. The resulting relative modelling variation is then transferred to the nominal SR selection. This requires the relative variation to be sufficiently stable under the relaxed requirements. Smoothing is applied only when the relaxed comparison still requires it; the two procedures are not automatically used together.

For the resonant selections, smoothing is used for the top parton-shower uncertainty in SR1b and for the matching and Wt interference uncertainties in VRTop. Relaxed selections without smoothing are used for the downward top FSR variation, h_{damp} and $t\bar{t}$ recoil in SR1b. For the non-resonant selection, the matching, h_{damp} and recoil variations use both a relaxed selection and smoothing. The $W + \text{jets}$ QCD-scale, top parton-shower and Wt interference variations use the relaxed selection without smoothing.

The dependence on the relaxed selection is checked using variations with sufficient statistics for a direct comparison. For the resonant top ISR variation, removing the jet-multiplicity requirement changes the transfer-factor ratio from 0.96 to 0.92, compared with an MC statistical uncertainty of 0.29 in that study. The smoothing procedure is also checked for the top matching variation in a relaxed resonant VRTop, using $N_b \geq 2$ and removing the jet-multiplicity requirement. The fractional changes before and after smoothing are 0.27 ± 0.10 and 0.15 ± 0.12 , respectively, which are consistent within their MC statistical uncertainties. These checks support the procedures in the tested regions, without establishing exact independence from every relaxed selection.

9.2.8 Summary and treatment of the variations

Table 9.1 summarises the evaluated theory variations for the SR1b transfer factors. The entries are changes in the relevant background component, before symmetrisation, and are not uncertainties in the total fitted background. In particular, a large fractional variation of the top-quark component need not produce an equally large change in the total background, which is dominated by $W + \text{jets}$.

Table 9.1 Fractional changes δ_b^R in the CR-to-SR1b transfer factors, expressed as percentages, after the smoothing or relaxed-selection treatment where used and before symmetrisation. For paired variations, the upward-parameter result is followed by the downward-parameter result in parentheses; their signs need not be opposite. A single entry denotes the signed change for an alternative prediction. The 30% heavy-flavour uncertainty and the uncertainties in minor-background normalisations are separate from these entries.

Source	SR1b-Res	SR1b-NonRes
W + jets QCD scales	+10 (−17)	+14 (−16)
W + jets PDF	< 0.1 in magnitude	
W + jets PDF α_s	−2 (−0.1)	+3 (−3)
W + jets electroweak correction	+4	+1
Top NLO matching	−6	−15
Top parton shower	+16	−1
Top factorisation scale	+12 (−9)	−5 (+5)
Top renormalisation scale	+15 (−19)	−1 (+2)
Top ISR	−4 (−1)	+16 (−13)
Top FSR	+19 (−24)	+1 (−25)
Top PDF	−5.1	−2.5
Top PDF α_s	+1 (−1)	−1 (+0.2)
Top h_{damp}	+9	−7
$t\bar{t}$ NNLO reweighting	−5	−0.3
$t\bar{t}$ shower recoil	−8	−13
Wt diagram removal/subtraction	−10	+3

Most paired variations are symmetrised to reduce sensitivity to MC fluctuations in sparsely populated regions. Depending on the source, the symmetric size is obtained from half the difference between the upward and downward shifts, or from the average of their absolute magnitudes. JER, soft-term and $t\bar{t}$ NNLO variations retain their dedicated treatment rather than being subjected to this general prescription. The raw values in Table 9.1 should therefore be distinguished from the nuisance-parameter variations used by the fit.

Finite MC sample sizes introduce a further statistical uncertainty in the predicted yields, separate from the modelling uncertainties above. It remains present after smoothing a modelling variation, since smoothing does not increase the number of simulated events in the nominal prediction. The fit includes dedicated parameters for the MC statistical uncertainty in each bin, as described in Chapter 10. The final impact of each uncertainty depends on the fitted background normalisations and correlations; it is assessed together with the statistical results in that chapter.

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